

TD Learning of state values:

essence: solve Bellman Equation without model.

→ v
Given $\pi: \{(s_t, r_{t+1}, s_{t+1})\}_t$

Algo (TD Learning)

$$\begin{cases} V_{t+1}(s_t) = v_t(s_t) - \alpha_t(s_t) [\overbrace{v_t(s_t) - (r_{t+1} + \gamma V_t(s_{t+1}))}^{\text{TD error } \delta_t}] \\ V_{t+1}(s) = V_t(s), \quad \forall s \neq s_t \end{cases}$$

TD target $\bar{v}_t$

$$V_t(s_t) \rightarrow V_\pi(s_t)$$

Property: ① $|V_{t+1}(s_t) - \bar{v}_t| \leq |V_t(s_t) - \bar{v}_t|$ closer to target
② $\delta_t \rightarrow 0$.

Sarsa (state-action-reward-state-action)

Algo (Sarsa)

Given $\pi \rightarrow \{(s_t, a_t, r_{t+1}, s_{t+1}, a_{t+1})\}_t$

$$\begin{cases} Q_{t+1}(s_t, a_t) = Q_t(s_t, a_t) - \alpha_t(s_t, a_t) [\underbrace{Q_t(s_t, a_t) - (r_{t+1} + \gamma Q_t(s_{t+1}, a_{t+1}))}_{\delta_t}] \\ Q_{t+1}(s, a) = Q_t(s, a), \quad \forall (s, a) \neq (s_t, a_t). \end{cases} *$$

$$Q_t(s_t, a_t) \rightarrow Q_\pi(s_t, a_t)$$

Process (1-step)

if $s_t \neq \text{target state}$, do

① get experience $(s_t, a_t, r_{t+1}, s_{t+1}, a_{t+1})$ by

$$s_t \xrightarrow{\pi_t(s_t)} a_t \longrightarrow r_{t+1}, s_{t+1} \xrightarrow{\pi_t(s_{t+1})} a_{t+1}$$

② update Q -value:

*

+ ③ update policy (π):

$$\pi_{t+1}(a|s_t) = \begin{cases} 1 - \frac{\epsilon}{|A|} (|A| - 1), & \text{if } a = \arg\max_a Q_{t+1}(s_t, a) \\ \frac{\epsilon}{|A|}, & \text{else} \end{cases}$$

Algo (Expected Sarsa)

$$Q_{t+1}(s_t, a_t) = Q_t(s_t, a_t) - \alpha_t(s_t, a_t) [Q_t(s_t, a_t) - (r_{t+1} + \gamma E[Q_t(s_{t+1}, A)])]$$

where

$$\hat{Q}_{t+1}(S_t, a_t) = \hat{Q}_t(S_t, a_t) - \alpha_t(S_t, a_t) [\hat{Q}_t(S_t, a_t) - (r_{t+1} + \gamma E[\hat{Q}_t(S_{t+1}, A)])]$$

where

$$E[\hat{Q}_t(S_{t+1}, A)] = V_t(S_{t+1})$$

no need for a_{t+1}

$$\hat{Q}_\pi(s, a) = E[G_t | S_t = s, A_t = a]$$

G_t :

$$\text{Sarsa} \leftarrow G_t^{(1)} = r_{t+1} + \gamma \hat{Q}_\pi(S_{t+1}, A_{t+1})$$

$$G_t^{(2)} = r_{t+1} + \gamma r_{t+2} + \gamma^2 \hat{Q}_\pi(S_{t+2}, A_{t+2})$$

$\vdots$

$$\text{n-step Sarsa} \leftarrow G_t^{(n)} = r_{t+1} + \gamma r_{t+2} + \dots + \gamma^{n-1} r_{t+n} + \gamma^n \hat{Q}_\pi(S_{t+n}, A_{t+n})$$

$\vdots$

$$\text{MC} \leftarrow G_t^{(\infty)} = r_{t+1} + \gamma r_{t+2} + \dots$$

$$n\text{-Bellman: } \hat{Q}_\pi(s, a) = E[G_t^{(n)} | s, a] = E[r_{t+1} + \gamma r_{t+2} + \dots + \gamma^{n-1} r_{t+n} + \gamma^n \hat{Q}_\pi(S_{t+n}, A_{t+n}) | s, a]$$

Algo (n-step Sarsa)

$$\hat{Q}_{t+1}(S_t, a_t) = \hat{Q}_t(S_t, a_t) - \alpha_t(S_t, a_t) [r_{t+1} + \dots + \gamma^{n-1} r_{t+n} + \gamma^n \hat{Q}_t(S_{t+n}, a_{t+n}) - \hat{Q}_t(S_t, a_t)]$$

use $(S_t, a_t, r_{t+1}, S_{t+1}, a_{t+1}, \dots, r_{t+n}, S_{t+n}, a_{t+n})$

$n=1$: Sarsa $n \rightarrow \infty$: MC

Q-Learning

solve a Bellman optimality Equation (BOE):

$$Q(s, a) = E[r_{t+1} + \gamma \max_a Q(S_{t+1}, a) | S_t = s, A_t = a], \quad \forall s, a.$$

Algo (Q-Learning)

Given $\pi \rightarrow \{(S_t, a_t, r_{t+1}, S_{t+1})\}_t$

$$\begin{aligned} \hat{Q}_{t+1}(S_t, a_t) &= \hat{Q}_t(S_t, a_t) - \alpha_t(S_t, a_t) [\hat{Q}_t(S_t, a_t) - (r_{t+1} + \gamma \max_{a \in A(S_{t+1})} \hat{Q}_t(S_{t+1}, a))] \\ \hat{Q}_{t+1}(s, a) &= \hat{Q}_t(s, a), \quad \forall (s, a) \neq (S_t, a_t). \end{aligned}$$

Off-policy v.s. on-policy

{ off-p: behavior-p $\neq$ target-p $\rightarrow \pi^*$: Q-learning.
on-p: behavior-p = target-p : Sarsa / MC

Q-Learning Pseudocode:

① on-policy

$$\pi_t \rightarrow (s_t, a_t, r_{t+1}, s_{t+1})$$

update q -value:

update policy: *behaviour = target*

$$\pi_{t+1}(a|s_t) = \begin{cases} 1 - \frac{\epsilon}{|A|} (|A| - 1), & \text{if } a = \operatorname{argmax}_a q_{t+1}(s_t, a) \\ \frac{\epsilon}{|A|}, & \text{else} \end{cases}$$



② off-policy

$$\pi_b \rightarrow \{s_0, a_0, r_1, s_1, a_1, r_2, \dots\}$$

update q -value:

update *target* policy:

$$\pi_{T,t+1}(a|s_t) = \begin{cases} 1, & \text{if } a = \operatorname{argmax}_a q_{t+1}(s_t, a) \\ 0, & \text{else.} \end{cases}$$